

Study title: The Role of Grassroots Innovation in Ukraine's Development of Unpiloted Systems

Context: On Feb. 22, 2014, Ukrainian protesters in Kyiv ousted the country's pro-Russian president, Viktor Yanukovich. In response, Russia annexed Crimea on Feb. 27 and helped form pro-Russian separatist paramilitaries in Donbas that began occupying eastern Ukraine in March. In April, Ukraine deployed its military forces to counter the Russian-backed separatists and stop their advance. This deployment marked the Ukrainian military's very first involvement in active warfare since the country's formation (in 1991). For eight years, the conflict remained confined to parts of eastern Ukraine. In the meantime, Ukraine worked with NATO countries to build its military capability. On Feb. 24, 2022, Russia escalated the conflict with an unsuccessful attempt to capture Kyiv, which turned into full-scale warfare between the two countries as Russia persisted with its attempt to occupy Ukraine. In response, Ukrainian society mobilized and began supporting its military. Four years into the full-scale warfare, military experts began to observe increasing success in Ukraine's ability to defend itself using unpiloted systems.

Objective: Describe how grassroots innovation contributed to Ukraine's development of unpiloted systems to defend against Russia's invasion, including the factors that drove and hindered the innovation, and how the innovation was scaled up and how it contributed to Ukraine's overall success in resisting Russia's invasion.

Study design: Single explanatory case study

Data collection: We will gather the digital versions of the Institute for the Study of War's daily war briefs. We will also review the historical record on social media and elsewhere to identify and document key events related to Ukraine's development of unpiloted systems. For each event, we will rely on multiple sources to determine the who, what, when, why, and where. We will also apply snowball sampling to identify Ukrainian and foreign civilians, soldiers, businesspersons, and government officials with firsthand experience in Ukraine's development of unpiloted systems, and interview them to gather firsthand explanations of (identified and not yet identified) events. In all interview questions, we will investigate the who, what, when, why, where, and how in relation to specific individual and collective, public and private activities and events. The interview questions will be semi-structured, aimed at gathering the following information:

- How did Ukraine's development of unpiloted systems evolve before your involvement? (This question is not applicable, if the person believes they were involved from the very start.)
- What was your involvement in Ukraine's development of unpiloted systems? What else was going on with Ukraine's development of unpiloted systems, in addition to your involvement?
- How did Ukraine's development of unpiloted systems evolve after you ended your involvement? (This question is not applicable, if the person is still involved.)

Analysis: We will use Natural Language Processing to analyze the Institute for the Study of War's daily war briefs, with the aim of identifying patterns in term (e.g., drone) use and frequency. We will use qualitative data analysis software to analyze the interviews and link information in them to specific events. We will then apply process tracing to identify the causal mechanisms linking events to the development of unpiloted systems. Finally, we will apply the Socio-Technical Transitions Framework to describe the four phases of Ukraine's development of unpiloted systems and the role of grassroots innovation.